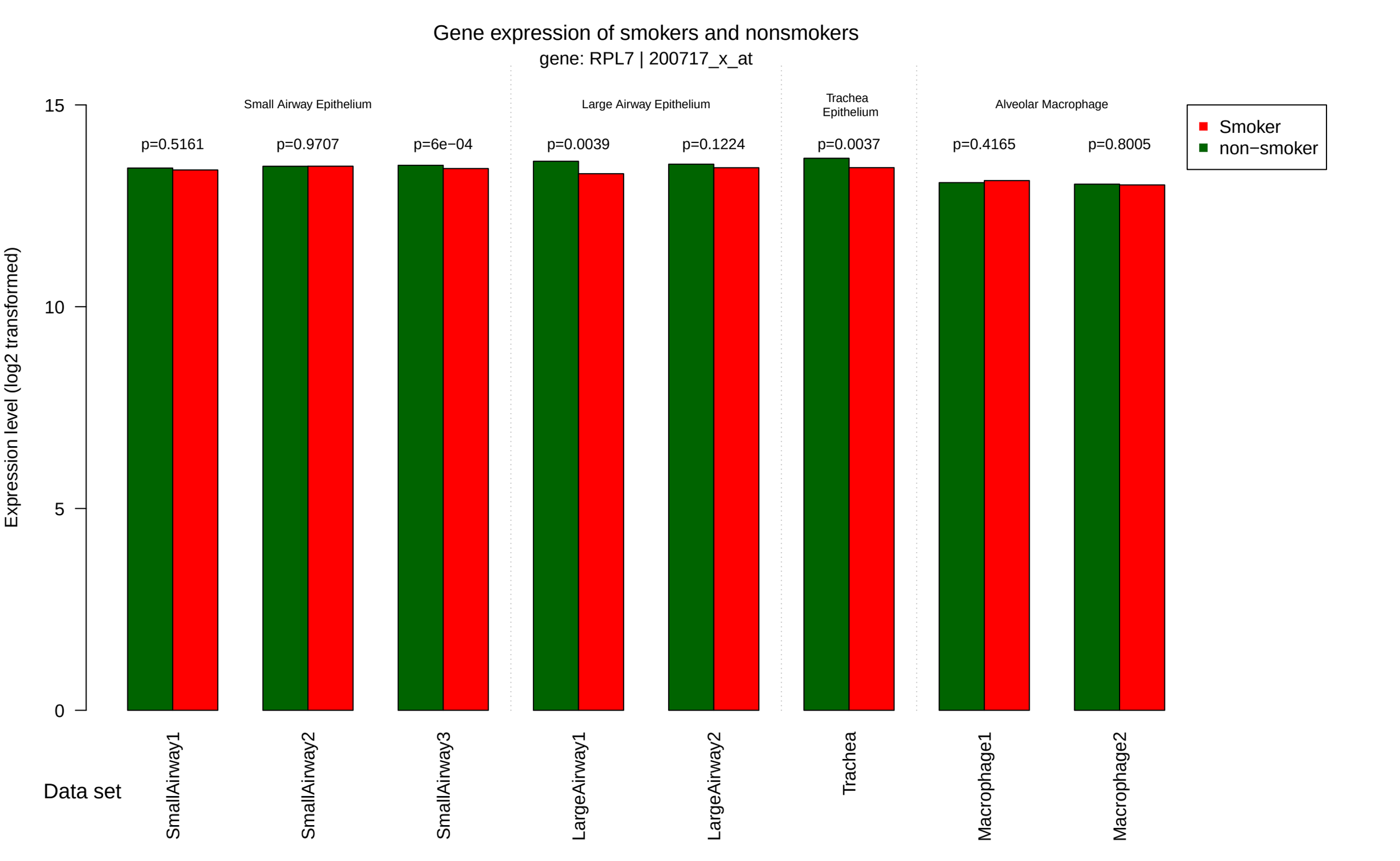
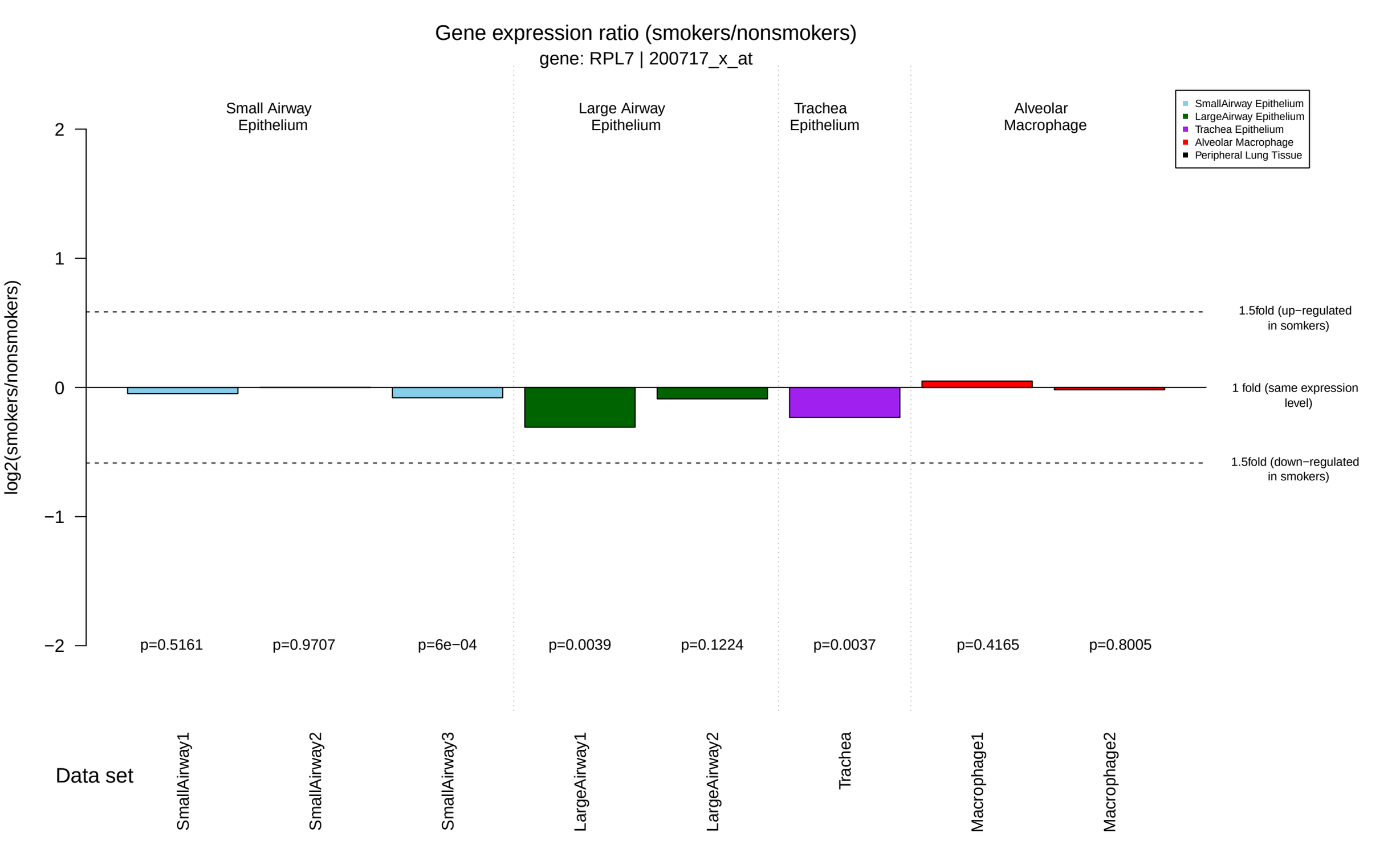
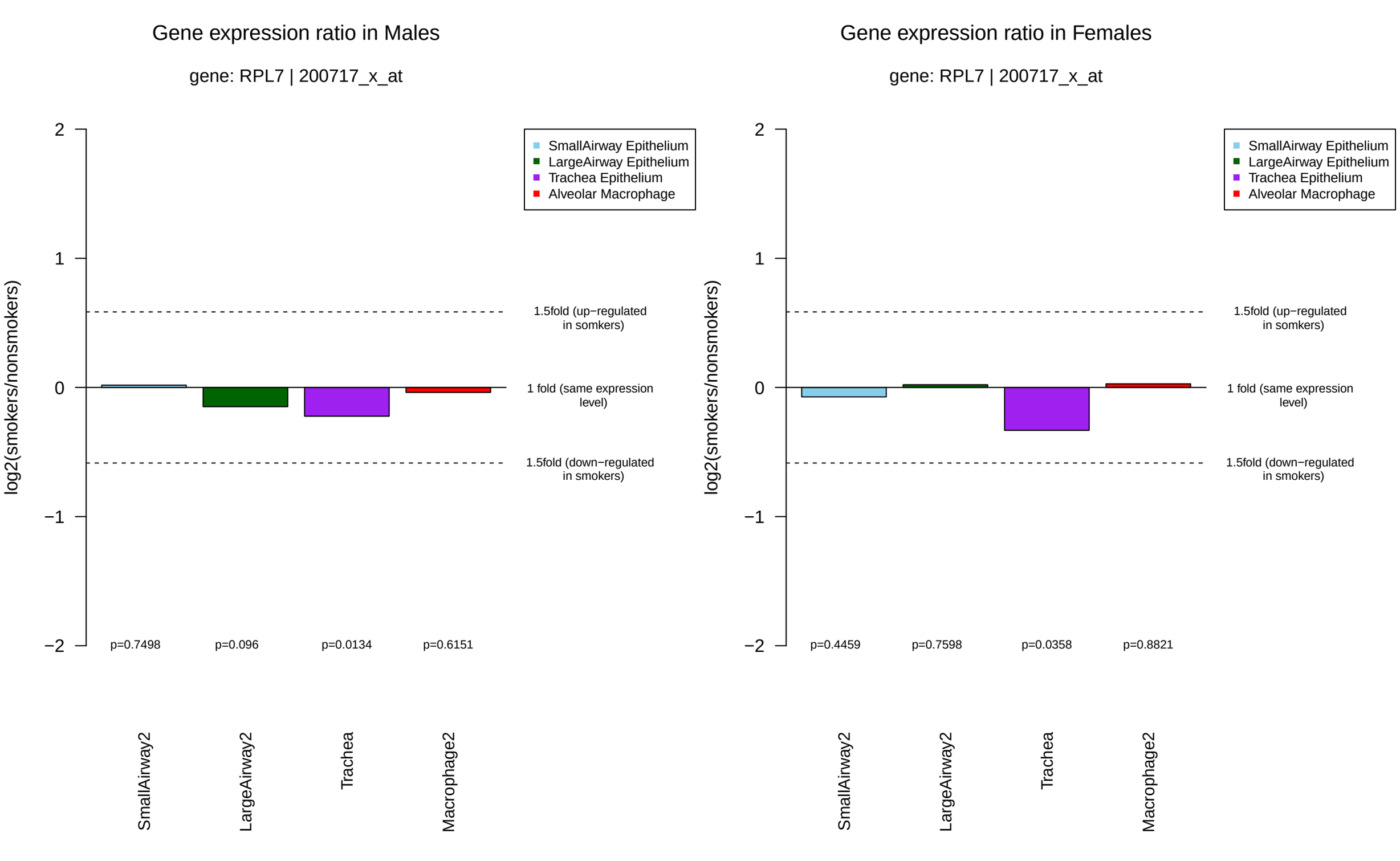
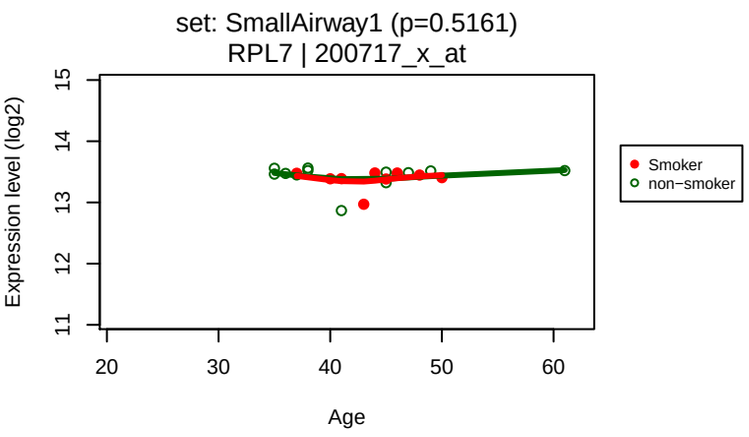
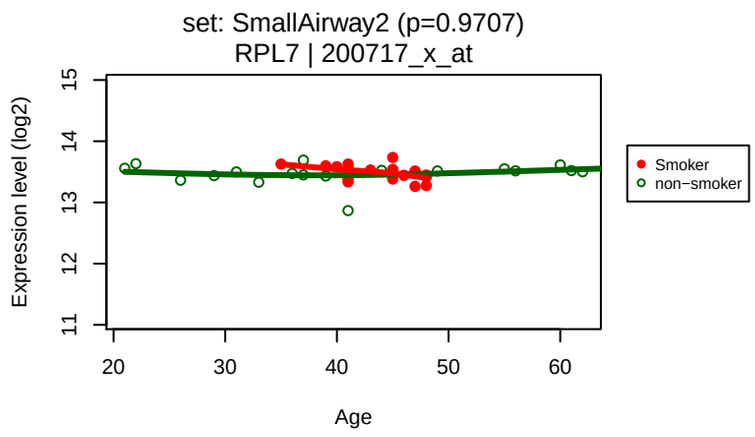
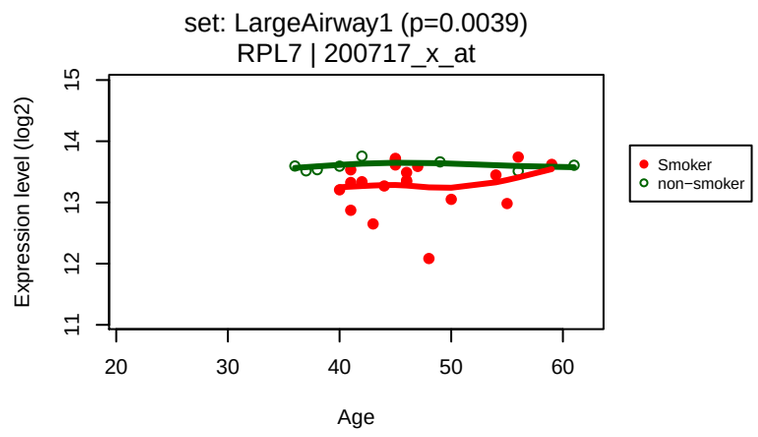

Gene expression of smokers and nonsmokers
gene: RPL7 | 200717_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5161
p=0.9707
p=6e−04
p=0.0039
p=0.1224
p=0.0037
p=0.4165
p=0.8005
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

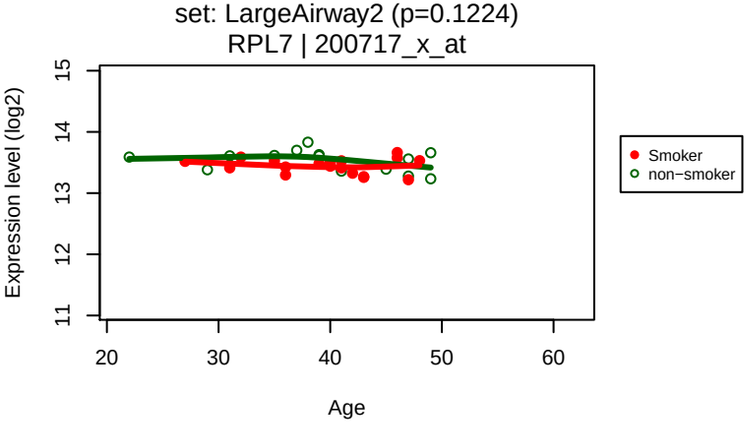

Gene expression ratio (smokers/nonsmokers)
gene: RPL7 | 200717_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5161
p=0.9707
p=6e−04
p=0.0039
p=0.1224
p=0.0037
p=0.4165
p=0.8005
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

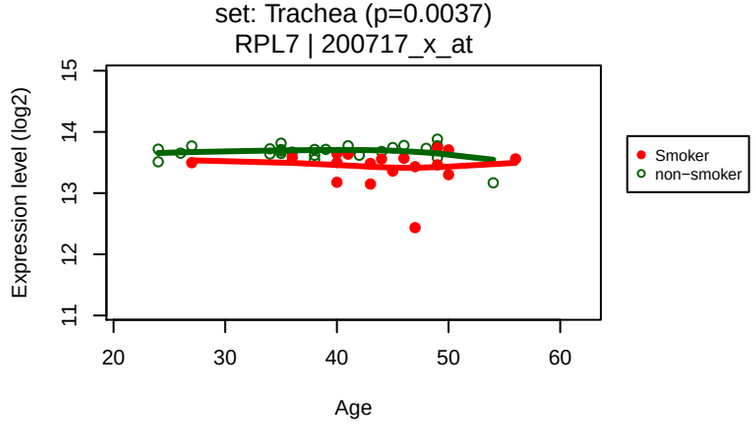

Gene expression ratio in Males
gene: RPL7 | 200717_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7498
p=0.096
p=0.0134
p=0.6151
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL7 | 200717_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4459
p=0.7598
p=0.0358
p=0.8821
SmallAirway2
LargeAirway2
Trachea
Macrophage2

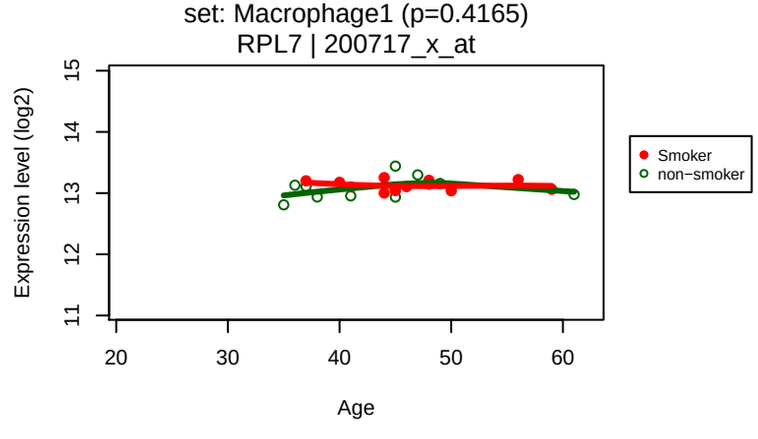

set: SmallAirway1 (p=0.5161)
RPL7 | 200717_x_at
Expression level (log2)
Age
Smoker
non−smoker

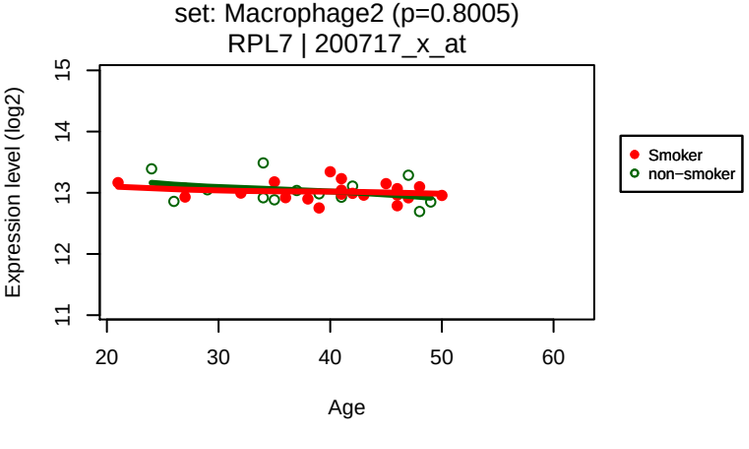

set: SmallAirway2 (p=0.9707)
RPL7 | 200717_x_at
Expression level (log2)
Age
Smoker
non−smoker

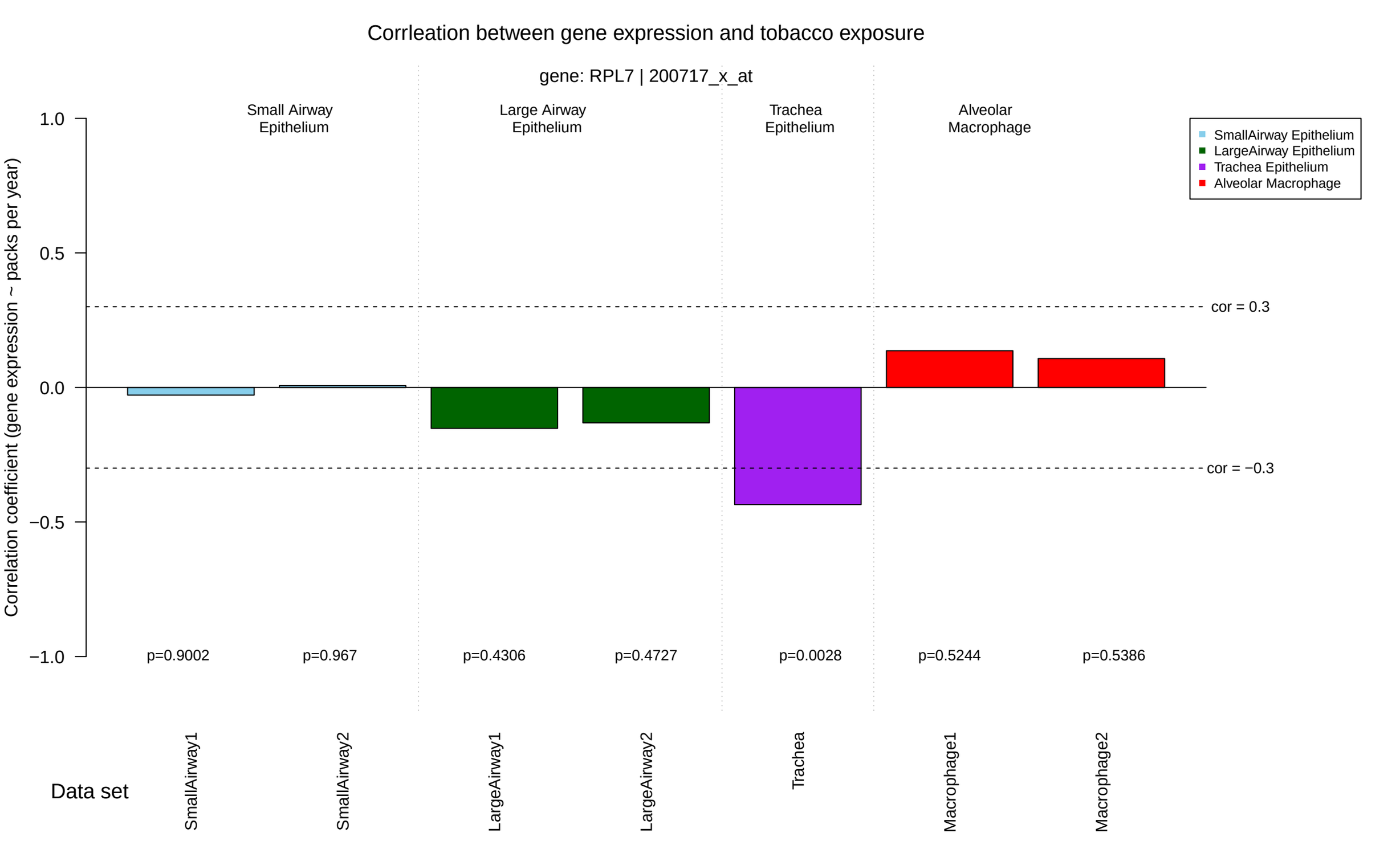

set: LargeAirway1 (p=0.0039)
RPL7 | 200717_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1224)
RPL7 | 200717_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0037)
RPL7 | 200717_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4165)
RPL7 | 200717_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8005)
RPL7 | 200717_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL7 | 200717_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9002
p=0.967
p=0.4306
p=0.4727
p=0.0028
p=0.5244
p=0.5386
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2